

Counting cards and other things

Rearranging letters

How many distinct ways could you rearrange the letters of the word “Mississippi”? Practice explaining! What general counting principle are you using? How does that principle help you count the rearrangements of these letters?

Bridge

The game of bridge is played with a standard 52-card deck. There are four players. Each player gets 13 cards. We call the 13 cards the players “hand.”

How many possible hands are there in the game of bridge?

How many possible ways are there to split the 52 cards in the deck into four 13-card hands for the four players?

How many bridge hands are there with four hearts cards, five spades cards, three clubs cards, and one diamonds card?

Poker

Poker is also played with a standard 52-card deck. A player gets five cards from the deck. These five cards are the player's "hand."

How many possible hands are there in poker?

Most cards in a standard deck have numbers (2, 3, 4, ..., 10). Poker players assign numbers to the other cards as well. They think of an ace (A) as a 1, a jack (J) as an 11, a queen (Q) as a 12, and a king (K) as a 13.

A "straight" is a poker hand where the values of the cards are consecutive (no gaps). For example, a poker hand with ace of hearts (1), two of clubs (2), three of clubs (3), four of diamonds (4), and five of spades (5) is a straight. Another straight is eight of spades (8), nine of hearts (9), ten of hearts (10), jack of diamonds (11), queen of spades (12).

How many straights are there where the lowest card is the ace of hearts?

What is the highest lowest card that could be in a straight?

How many poker hands are straights?